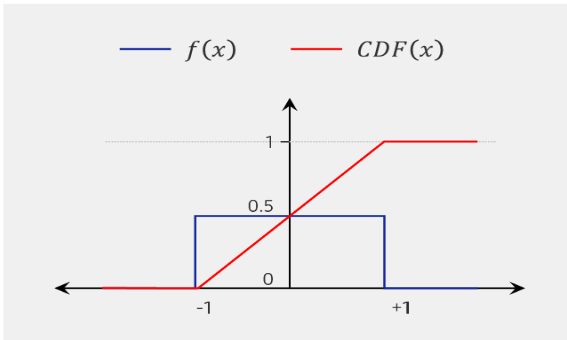
1)Continous probability density

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import uniform

# Parameters
a = -1
b = 1
width = b - a

# x range
x = np.linspace(-2, 2, 400)

# PDF and CDF
pdf = uniform.pdf(x, loc=a, scale=width)
cdf = uniform.cdf(x, loc=a, scale=width)

# Plot
plt.figure(figsize=(8, 5))
plt.plot(x, pdf, color='blue', linewidth=2, label="f(x)  (PDF)")
plt.plot(x, cdf, color='red', linewidth=2, label="CDF(x)")

# Axes lines
plt.axhline(0, color='black', linewidth=1)
plt.axvline(0, color='black', linewidth=1)

# Labels
plt.title("Uniform Distribution: a = -1, b = +1")
plt.xlabel("x")
plt.ylabel("Probability / Density")
plt.legend()

plt.grid(False)
plt.show()
```

Uniform random variable and probability density

Ex $a = -1, b = +1$

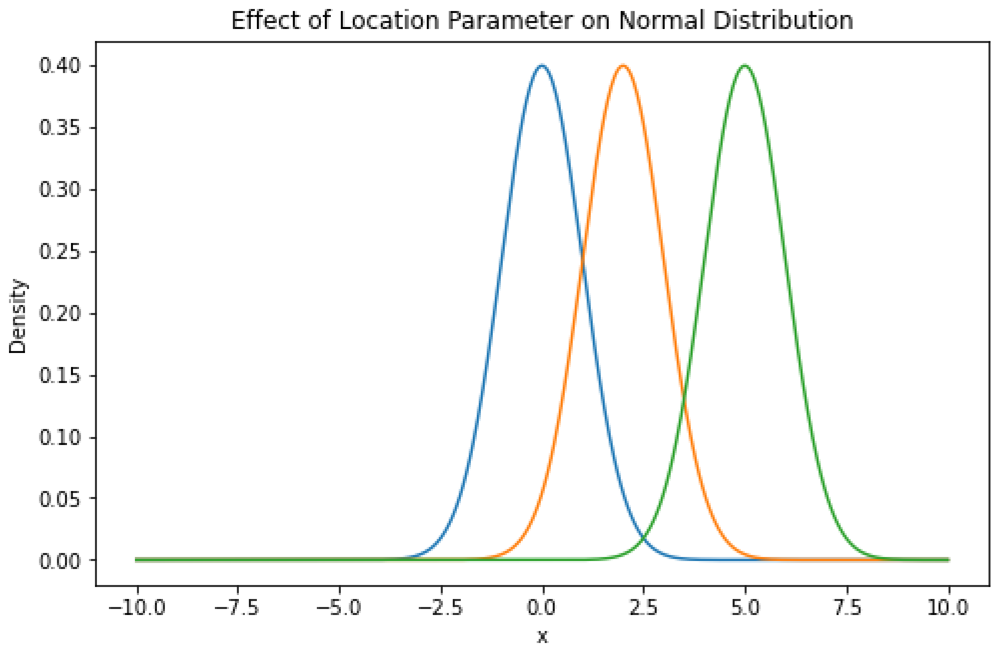


This code produces a plot with:

- **A blue rectangle** from x = −1 to +1 with height 0.5 (PDF)
- **A red straight-line CDF** increasing from 0 ⟶ 1 between x = −1 and +1
- Flat regions for x < −1 and x > 1

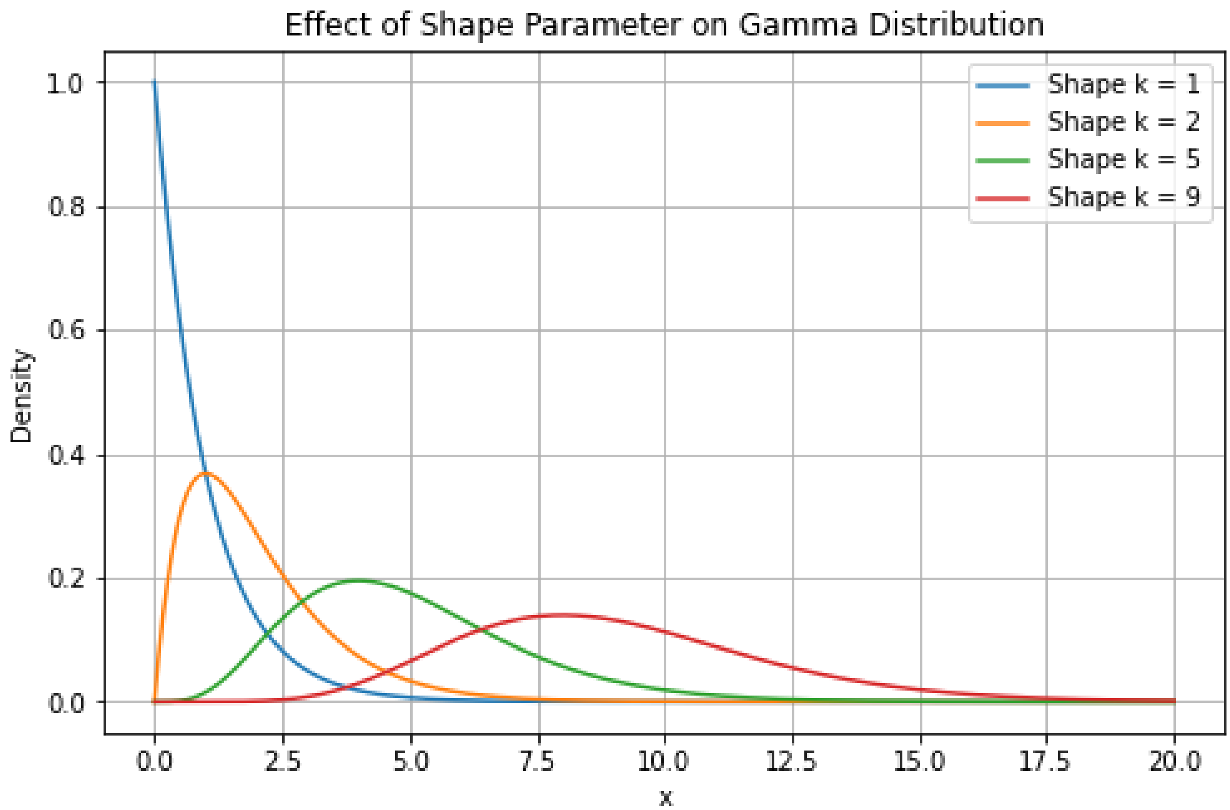
2) location parameter

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

x = np.linspace(-10, 10, 500)

# different location parameters
mu_values = [0, 2, 5]
sigma = 1

plt.figure(figsize=(8, 5))

for mu in mu_values:
   plt.plot(x, norm.pdf(x, loc=mu, scale=sigma), label=f"μ = {mu}")

plt.title("Effect of Location Parameter on Normal Distribution")
plt.xlabel("x")
plt.ylabel("Density")
plt
```



3)Shape parameter

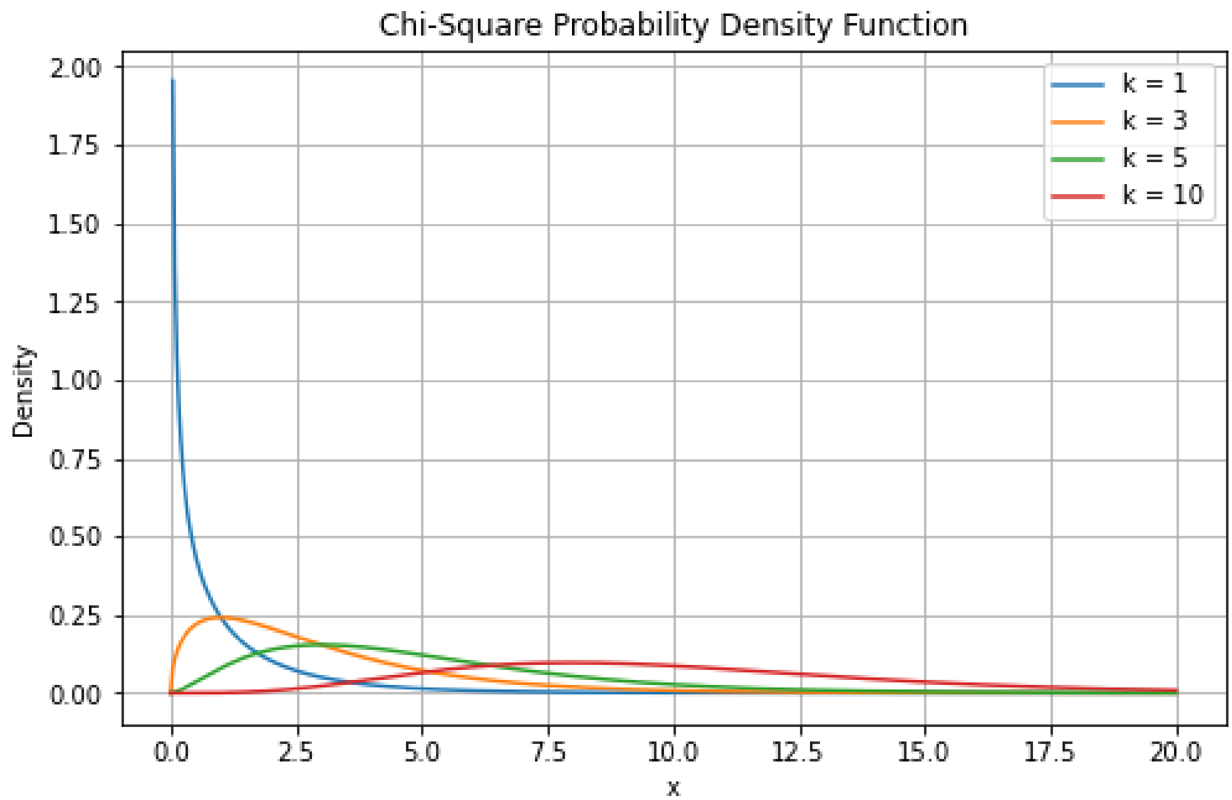
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gamma

# x-axis values
x = np.linspace(0, 20, 500)

# different shape parameters (k values)
shape_params = [1, 2, 5, 9]

plt.figure(figsize=(8, 5))

for k in shape_params:
    plt.plot(x, gamma.pdf(x, a=k), label=f"Shape k = {k}")

plt.title("Effect of Shape Parameter on Gamma Distribution")
plt.xlabel("x")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.show()
```



4)Chi-Square Random Variable

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2

x = np.linspace(0, 20, 500)

k_values = [1, 3, 5, 10]   # degrees of freedom

plt.figure(figsize=(8, 5))

for k in k_values:
    plt.plot(x, chi2.pdf(x, df=k), label=f"k = {k}")

plt.title("Chi-Square Probability Density Function")
plt.xlabel("x")
plt.ylabel("Density")
```

```
plt.legend()
plt.grid(True)
plt.show()
```



5)Chi-square PDF

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2

k = 4  # degrees of freedom
x = np.linspace(0, 20, 500)
pdf_vals = chi2.pdf(x, df=k)

plt.plot(x, pdf_vals)
plt.xlabel("x")
plt.ylabel("PDF")
plt.title(f"Chi-Square PDF (df = {k})")
plt.grid(True)
plt.show()
```

6) Chi-square CDF

```
cdf_vals = chi2.cdf(x, df=k)

plt.plot(x, cdf_vals)
plt.xlabel("x")
plt.ylabel("CDF")
plt.title(f"Chi-Square CDF (df = {k})")
plt.grid(True)
plt.show()
```

7)chi-square goodness of fit example

```
import numpy as np
from scipy.stats import chisquare

observed = np.array([25, 30, 20, 25])
expected = np.array([25, 25, 25, 25])

chi_stat, p_val = chisquare(observed, expected)

print("Chi-square statistic:", chi_stat)
print("p-value:", p_val)
```

8) Python Code for Student-t Distribution

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import t

df = 5
x = np.linspace(-5, 5, 500)
pdf_vals = t.pdf(x, df=df)

plt.plot(x, pdf_vals)
plt.xlabel("x")
plt.ylabel("PDF")
plt.title(f"Student t PDF (df={df})")
plt.grid(True)
```

```
plt.show()
```

9)PLOT CDF-cumulative distribution function

```
cdf_vals = t.cdf(x, df=df)

plt.plot(x, cdf_vals)
plt.xlabel("x")
plt.ylabel("CDF")
plt.title(f"Student t CDF (df={df})")
plt.grid(True)
plt.show()
```

10) Python Example: Zero Correlation but Strong Curve

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import pearsonr

x = np.linspace(-3, 3, 200)
y = x**2  # nonlinear relationship

corr, _ = pearsonr(x, y)
print("Correlation =", corr)

plt.scatter(x, y)
plt.title("Non-linear relationship (Correlation ≈ 0)")
plt.show()
```

11)Strong positive and weak positive correlation example

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import pearsonr

# -----------------------------
# Strong Positive Correlation
# -----------------------------
np.random.seed(0)
x_strong = np.linspace(1, 10, 50)
```

```
y_strong = 5 * x_strong + np.random.normal(0, 3, size=50)  # strong linear trend + small noise

# correlation
corr_strong, _ = pearsonr(x_strong, y_strong)
print("Strong positive correlation (r):", corr_strong)

# plot
plt.scatter(x_strong, y_strong)
plt.xlabel("X (Strong)")
plt.ylabel("Y (Strong)")
plt.title("Strong Positive Correlation")
plt.grid(True)
plt.show()


# -----------------------------
# Weak Positive Correlation
# -----------------------------
np.random.seed(1)
x_weak = np.linspace(1, 10, 50)
y_weak = 1 * x_weak + np.random.normal(0, 8, size=50)  # small positive trend + large noise

# correlation
corr_weak, _ = pearsonr(x_weak, y_weak)
print("Weak positive correlation (r):", corr_weak)

# plot
plt.scatter(x_weak, y_weak)
plt.xlabel("X (Weak)")
plt.ylabel("Y (Weak)")
plt.title("Weak Positive Correlation")
plt.grid(True)
plt.show()
```